RaiseIt

Problem Statement

In lecture halls, students often hesitate to raise their hands due to fear of being overlooked or feeling awkward if not acknowledged. Professors, on the other hand, may struggle to identify and address every raised hand, leading to missed opportunities for engagement.





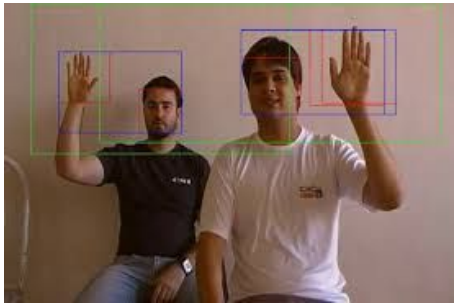


Solution

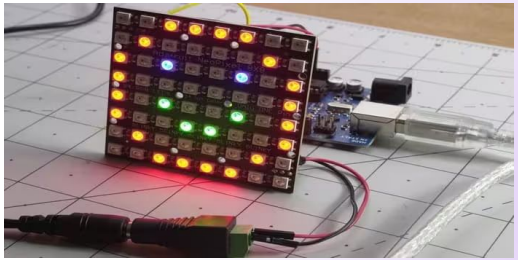
Our solution is to create an image recognition system that will identify when and where a raised hand is within a lecture which would then send a signal to the professor using a light grid UI/LED matrix at the front of the lecture

This can be done using:

- OpenCV(Identify the raised hands)
- TensorFlow(To send the signal to the light fixture)
- Raspberry Pi
- Webcam

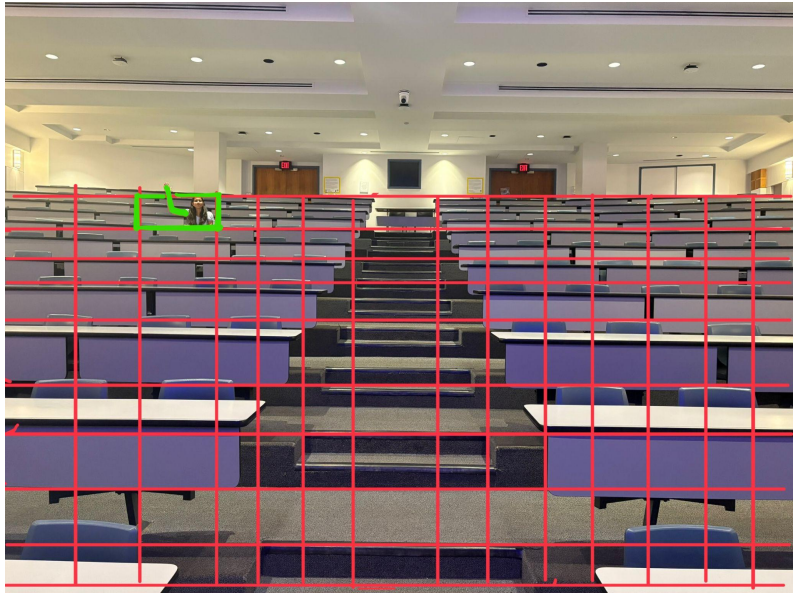


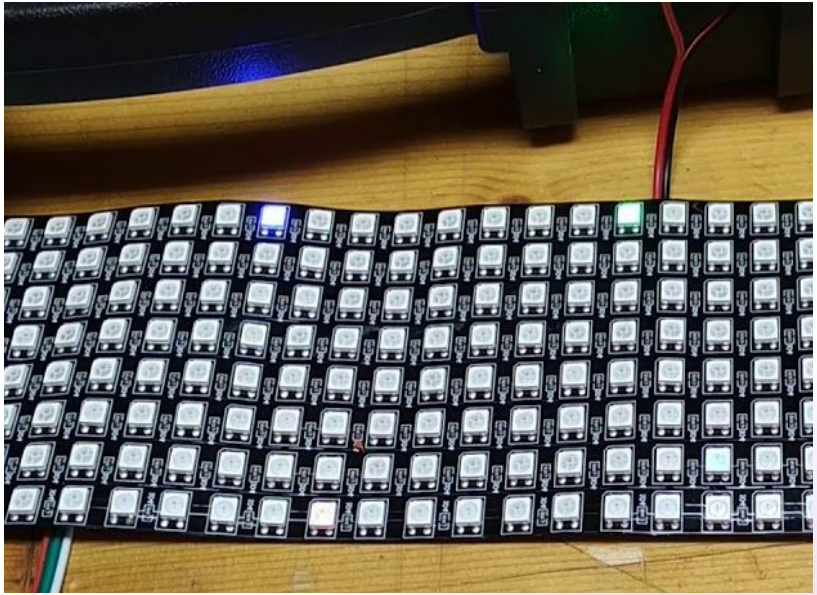




(Through OpenCV and mediapipe package)

Solution





Solution

For Professors



For Students



Competition and Competitive Advantages

Existing Solutions

- Manual hand-raising
- Clicker systems
- Digital platforms for questions

Each with limitations in real-time interaction and inclusivity.

RaiseIt Advantages

- Real-time detection and notification
- Non-intrusive integration into existing classroom setups

Enhanced student engagement by ensuring all participation attempts are acknowledged.

The Next Steps

Phase 1: Gage Interest & Identify Key Features

Phase 2: Prototype Development (Test in a real lecture hall). Collect feedback to refine the solution.

Phase 3: Pilot at Lehigh University. Analyze data, refine features, and assess overall effectiveness.

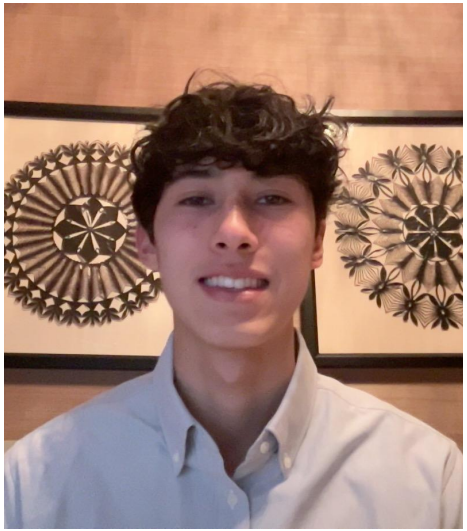
Phase 4: If successful, expand to other colleges

Use of Funds:

Prototype Development

Acquiring materials and software necessary to build a functional prototype.

The Team



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Thank You!

We are open to Questions now!

Do you have any questions?

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